# ZHUOYI(CARL) CUI

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## EDUCATION

CARNEGIE MELLON UNIVERSITY, TEPPER SCHOOL OF BUSINESS — Pittsburgh, PA
*Master of Science in Computational Finance – MSCF* — ***GRE Quant: 170/170*** — 08/25 - 12/26
- Coursework: Financial Data Science, Time Series Analysis, Financial Optimization, Stochastic Calculus

NANJING UNIVERSITY(Top 0.5% in Chinese College Entrance Exam) — Nanjing, China
*Bachelor of Economics in Financial Engineering* — ***GPA: 4.54/5.00; RANK: 2/38*** — 08/21 - 06/25
- Coursework: Applied Statistics, Numerical Methods, OOP(Python), Data Structures, Fixed Income

UNIVERSITY OF CALIFORNIA, BERKELEY — Berkeley, CA
Berkeley International Exchange Program — ***GPA:3.96/4.00*** — 01/23 - 12/23
- Coursework: Machine Learning, Deep Learning, Stochastic Processes, Econometrics, Asset Pricing

## EXPERIENCE

MORGAN STANLEY — New York, NY
*Quantitative Research Summer Associate* — 06/26 - 08/26
- **Signal Research:** Researched analyst forecast-revision signals incorporating analyst-level characteristics such as boldness and forecasting skill; engineered earnings-call casting features to capture management communication patterns using FactSet data; with validated signals achieving 1.5+ annualized Sharpe out of sample
- **Valuation Framework:** Built regression-based framework to identify mispricing between observed and fundamentals-implied valuation multiples using refined growth, durability, and reinvestment metrics; the composite signal delivered ~3× the average return and ~50% lower max drawdown versus baseline signals across multiple equity universes

SHANGHAI REDWALL TAIHE FUND MANAGEMENT COMPANY LIMITED — Shanghai, China
*Quantitative Research Intern, High-Frequency Group* — 11/24 - 04/25
- **Alpha Research:** Constructed minute-frequency alphas with tick-level order and trade data using Polars, topics include digit distribution abnormality, trade co-occurance, order cancellation, etc; incorporated into company's factor pool and achieving 3% out-of-sample performance improvement
- **Backtest Tool Development:** Built a minute-level factor backtesting framework with memory-mapped storage and GPU-based vectorized computation, running 2-year simulation across 5,000+ stocks in under 20 seconds; integrated block-bootstrap tests to correct hyperparameter level multiple testing and produced selection-adjusted confidence level

BEIJING PRISM PRIVATE FUND MANAGEMENT CO. LTD. — Beijing, China
*Quantitative Research Intern, Mid-Frequency Group* — 07/24 - 9/24
- **ML Factor Combination:** Adopted architectural modification and regularization techniques for factor combination, elevating baseline IC by 15%, while reducing the overall turnover rate by approximately 20% for Neural Net models
- **LLM Sentiment Analysis:** Utilized LLMs for sentiment analysis using 100K+ corporate related news data, applied LoRA based fine-tuning for financial domain adaptation, identify high risk stocks to adjust position allocations

## RESEARCH

*Research Assistant* | **University of Chicago, Booth School of Business**| *Led by Prof. Dacheng Xiu*
DEEP AUTOENCODERS FOR NONLINEAR FACTOR MODELS — 03/24 - 09/24
- **Statistical ML for factor Models:** Investigated the capability of autoencoders(AE) and supervised autoencoders(SAE) in non-linear factor model settings, focusing on feature extraction, factor augmented regression and matrix completion
- **Model Optimization:** Adjusted decoder architecture as implicit regularization; conducted hyper-parameter tuning for optimal performance, reducing reconstruction error by 40% relative to traditional non-parametric models such as PCA

*Research Assistant* | **University of Chicago, Booth School of Business** | *Led by Prof. Dacheng Xiu*
PREDICTIVE MODELING UNDER WEAK SIGNALS — 07/23 - 12/23
- **Regression Under Weak Signal Settings:** Examined performance of linear models(Ridge, LASSO and OLS) and non-linear models(RF, GBDT and NN) through error-bound analysis, monte-carlo simulation and empirical research on financial(cross-sectional equity return prediction) and economic(GDP growth prediction) datasets; gained insights on regression practices utilizing weak signals, mixed strength signals and regression residuals for robust prediction

## ADDITIONAL INFORMATION

**Interests:** Texas-Hold'em, Chinese Go(4-dan amateur), Basketball, Watching Classic Movies
**Programming Skills:** Python(Pandas, Polars, Numpy, Pytorch), Matlab, R, Git, SQL, Claude/Codex Terminal